

What do Transit and Radial Velocity reveal about exoplanets?

A first hands-on analysis of orbital periods and planet masses using the NASA Exoplanet Archive

I wanted to learn data analysis by working on a real question, not by following another generic tutorial. My plan was simple: spend an hour at night with a dataset, decide what analysis would actually make sense, look up concepts when I got stuck, and build the analysis myself. For this first project, I chose exoplanets because the data connects directly to the physics I am interested in. I compared planets discovered through the **Transit** and **Radial Velocity** methods and asked whether the planets represented by the two methods look different in orbital period and reported mass.

Why this project?

This was not meant to be a research paper or a claim of a new astronomical discovery. It was the first part of my data-analysis training: take a real scientific dataset, ask a focused question, clean it carefully, make defensible plots, and learn how easily a graph can become misleading when the underlying data are not understood.

The project became especially useful because the first version of the dataset looked straightforward but was not. There were tens of thousands of rows, yet only a few thousand distinct planets. Missing measurements were distributed very differently across the two discovery methods. The mass column also mixed actual masses with minimum-mass estimates. Each of those details changed how the analysis had to be designed.

The question I used

Do planets discovered using Transit and Radial Velocity have different observed orbital-period and mass distributions?

The word *observed* matters. This analysis compares what appears in the archive, not the complete underlying population of planets in the Galaxy.

A quick bit of context

The two methods do not look for planets in the same way.

Method	What is measured	What it naturally provides
Transit	A small drop in a star's brightness when a planet crosses in front of it	Orbital period and planet radius, provided the orbital alignment is favourable
Radial Velocity	The Doppler shift caused by a star moving toward and away from us under a planet's gravitational pull	Orbital period and a mass estimate, often $M \sin i$ when inclination is unknown

Because the methods respond to different signals, I expected some difference between the samples. The interesting part was finding out how large that difference was and deciding how much of it could be trusted.

The data and the first surprise

I used the [NASA Exoplanet Archive Planetary Systems table](#).

The table stores one row per planet per published reference. This means the same planet can appear several times with different reported parameter sets. The archive provides a `default_flag` that can be used to retain one preferred row per planet.

Table 1: What the dataset looked like before and after cleaning.

Quantity	Value	Why it mattered
Columns in the downloaded table	92	The table contains discovery, orbital, planet, stellar and reference information.
Rows after selecting Transit and Radial Velocity	38,754	This was much larger than the number of actual planets because repeated published parameter sets were included.
Unique planets across all discovery methods	6,319	A useful overall check on the downloaded table.
Unique planets in the Transit and RV subset	5,857	This became the population used for the main comparison.
Transit planets after <code>default_flag = 1</code>	4,666	One preferred row per Transit planet.
Radial Velocity planets after <code>default_flag = 1</code>	1,191	One preferred row per RV planet.

The difference between 38,754 rows and 5,857 unique planets was the most important cleaning discovery. Had I treated every row as an independent planet, well-studied planets would have been counted repeatedly and would have had more influence on the summary statistics and plots.

A mistake I caught

At first, I was analysing all rows. Checking the number of unique values in `p1_name` showed that the table contained repeated planets. I then filtered to `default_flag = 1`, which produced exactly 5,857 rows, one for every unique planet in the two-method subset.

How I made the analysis choices

I kept missing values as `NaN` rather than filling them with averages. Filling a missing mass or radius with a sample mean would create a measurement that was never observed. Instead, I created a separate usable subset for each question.

Table 2: Decision log: what I chose and why.

Part	Choice	Value or rule	Reason
Methods	Discovery methods	Transit and Radial Velocity	They were the two dominant methods and offered a physically meaningful comparison.
Duplicates	Preferred record	<code>default_flag = 1</code>	Prevented the same planet from being counted several times.

Part	Choice	Value or rule	Reason
Identity check	Planet name	pl_name	Used to compare total rows with the number of unique planets.
Missing data	Imputation	None	Replacing missing scientific measurements with averages would invent data.
Orbital period	Variable	pl_orbper, measured in days	Directly answers how quickly each planet completes an orbit.
Orbital period	Usable rows	Positive, non-missing values	Missing or non-positive periods cannot be used in a logarithmic physical comparison.
Orbital period	Main measure of centre	Median	The distribution has a long right tail, so the mean is strongly affected by very long-period planets.
Orbital period	Histogram range	10^{-1} to 10^4 days	Covers the displayed range from fractions of a day to ten thousand days.
Orbital period	Bins	50 logarithmically spaced bins	The values span several orders of magnitude. Linear bins would compress most planets into the left edge.
Histogram comparison	Vertical scale	Percentage within each method	The Transit and RV sample sizes were unequal, so raw counts would mainly show which sample was larger.
Planet property	Radius candidate	pl_rade	Transit directly measures radius, so radius initially looked like the obvious variable.
Planet property	Radius rejected	Only 31 RV radii were available	A radius comparison would represent the entire RV population using a tiny and highly selected subset.
Planet property	Mass selected	pl_bmasse, measured in Earth masses	It provided usable samples for both methods: 1,191 RV planets and 1,476 Transit planets.
Mass	Histogram range	10^{-2} to 10^4 Earth masses	Included the observed positive range, from 0.07 to more than 9,000 Earth masses.
Mass	Bins	50 logarithmically spaced bins	The mass values span more than five orders of magnitude.

Part	Choice	Value or rule	Reason
Mass	Provenance check	pl_bmassprov	The mass column mixes actual mass, $M \sin i$, and inclination-corrected estimates.
Mass valley	Interpretation	Described as apparent, not proven	A histogram alone cannot separate real planet physics from selection effects, follow-up bias or binning effects.

Analysis 1: orbital periods

The first comparison was orbital period. After keeping one preferred row per planet and removing unusable period values, the medians were:

RADIAL VELOCITY MEDIAN

307.88 days

TRANSIT MEDIAN

8.027 days

The typical RV discovery in this sample had an orbital period about 38 times longer than the typical Transit discovery.

I used the median rather than relying on the mean because a small number of very long-period planets pull the mean upward.

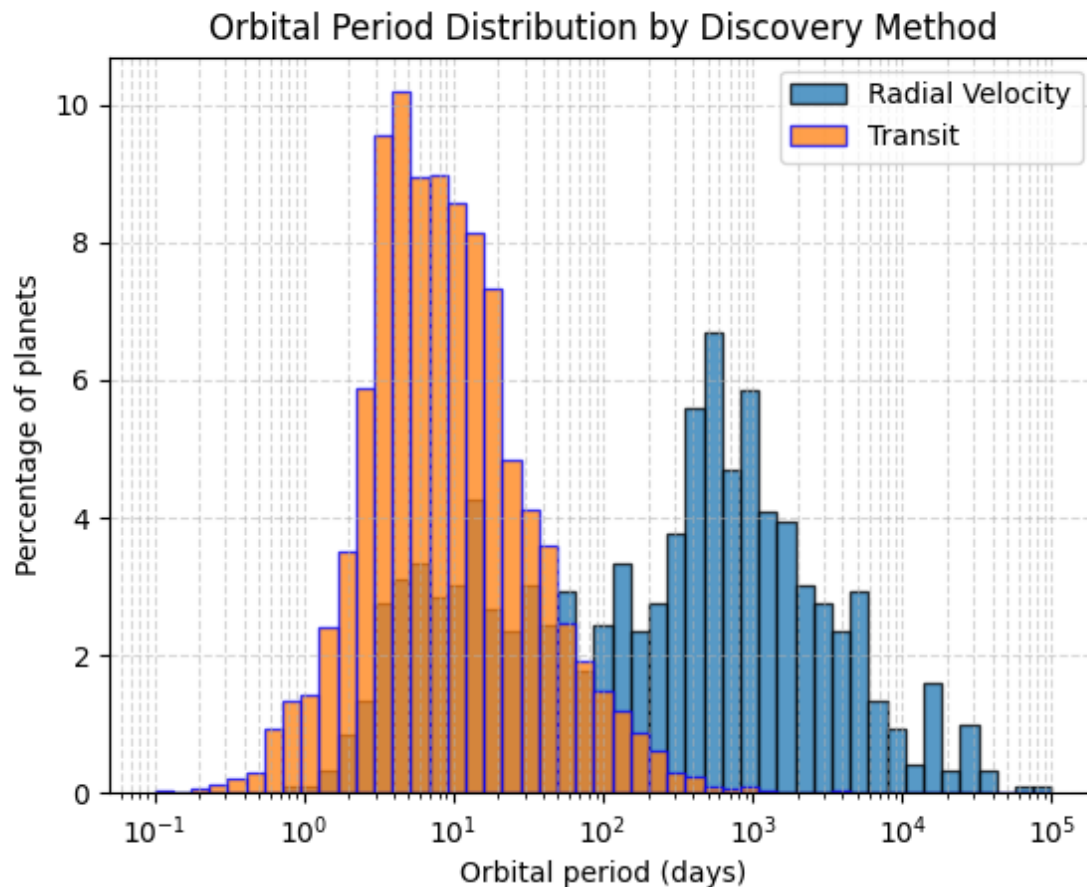


Figure 1: Orbital-period distributions, with each method normalised to its own percentage total. Both the bins and the horizontal axis are logarithmic.

What I see in the graph

Transit discoveries are strongly concentrated at short periods, particularly from a few days to a few tens of days. Radial Velocity discoveries cover a much wider range and have a substantial concentration at hundreds to thousands of days. There is still overlap, so the methods are not observing completely separate classes of planets.

This pattern is consistent with method-dependent selection. Short-period planets produce more transits during a limited observing campaign. RV surveys can recover longer periods when the same stars are followed for long enough.

That does not mean every short-period planet is a Transit planet or every long-period planet is an RV planet. It means the methods sample the underlying population differently.

Analysis 2: planet mass

Why I did not use radius

Radius had better coverage overall, but that headline number was misleading because the dataset was dominated by Transit discoveries.

Once I checked the methods separately, the imbalance became obvious.

Table 3: Measurement availability after keeping one default row per planet.

Method	Total	Radius values	Radius coverage	Mass values	Mass coverage
Radial Velocity	1,191	31	2.60%	1,191	100.00%
Transit	4,666	4,639	99.42%	1,476	31.63%
Combined	5,857	4,670	79.73%	2,667	45.54%

A cross-method radius comparison would therefore have been unfair. The RV side would contain only 31 planets.

Mass was the better exploratory choice because both methods had sizeable samples, although the Transit mass sample was still only about one-third of all Transit discoveries.

The mass statistics

Table 4: Reported mass statistics in Earth masses.

Statistic	Radial Velocity	Transit
Usable planets	1,191	1,476
Minimum	0.193	0.070
First quartile	14.994	8.900
Median	291.120	82.636
Mean	961.443	347.716
Third quartile	1,136.576	324.189
Maximum	9,333.031	8,899.195

Both groups are strongly right-skewed. Their means are much larger than their medians, and their maximum values sit far beyond the third quartile.

The median RV mass was about 3.5 times the median Transit mass in the measured sample.

One complication: “mass” did not always mean the same thing

The archive’s `p1_bmasse` column stores the best available mass estimate, but the provenance differs between planets.

Table 5: What kind of mass estimate was stored in `p1_bmasse`.

Provenance	RV count	RV share	Transit count	Transit share
Mass	235	19.73%	1,466	99.32%
$M \sin i$	951	79.85%	8	0.54%
$M \sin(i)/\sin(i)$	5	0.42%	2	0.14%
Total	1,191	100.00%	1,476	100.00%

Most RV entries are $M \sin i$, a minimum mass when inclination is unknown, while almost every Transit entry is labelled as mass.

I kept the archive's best available estimate for the main exploratory plot, but this means the two distributions are not perfectly like-for-like.

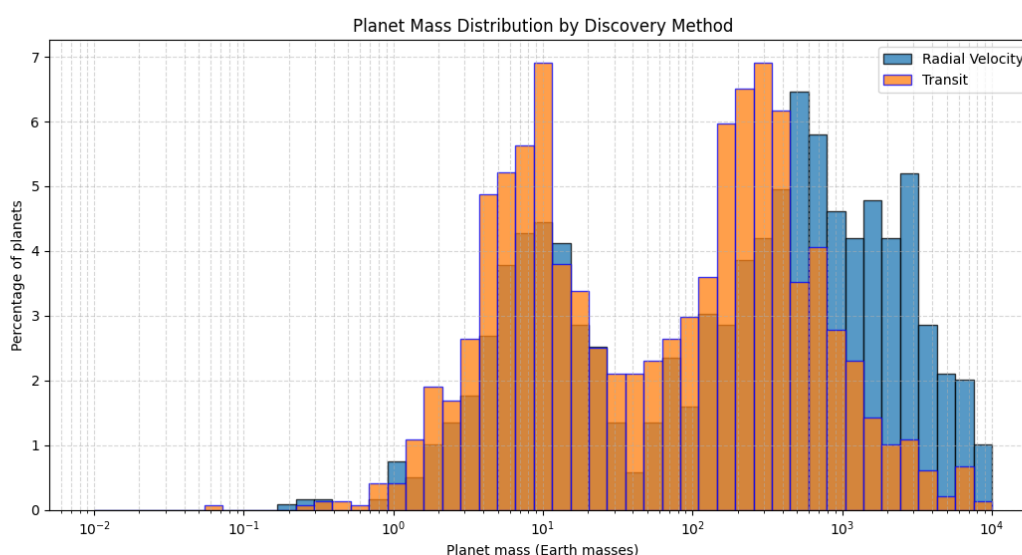


Figure 2: Reported planet-mass distributions. Each method is normalised separately, and both the bins and horizontal axis are logarithmic.

What I see in the graph

The RV distribution is shifted toward larger masses.

Transit shows a strong lower-mass concentration around a few to a few tens of Earth masses, while both methods contain many giant planets between roughly 10^2 and 10^3 Earth masses.

RV has a stronger tail into several thousand Earth masses.

The part that caught my eye: the valley

The mass histogram has a shape I did not expect. There appears to be one concentration at lower masses, another among giant planets, and a valley between them.

It is tempting to immediately give that valley a physical explanation, but this analysis cannot do that.

The feature could be related to planet formation, but it could also be produced or exaggerated by:

- detection sensitivity,
- which Transit planets receive successful mass follow-up,
- the mixture of true mass and $M \sin i$,
- or the exact choice of histogram bins.

The careful conclusion

The graph shows an **apparent** valley or bimodal pattern. It is a question generated by the analysis, not an answer established by it.

That was probably the most useful moment in the project. Data analysis was no longer just “make a graph and explain it.”

The graph created a new question that would need better completeness corrections, robustness checks, domain knowledge, and possibly guidance from an exoplanet researcher.

What I learned from this project

1. A row is not automatically an independent observation. Understanding what one row represents came before any statistics.
2. Missing values should be handled according to the question. Removing every row with any missing field would have thrown away useful information.
3. Overall coverage can hide severe subgroup imbalance. Radius looked well measured until I separated Transit and RV.
4. Normalising a histogram is different from standardising the underlying values. I needed percentages, not z-scores.
5. Logarithmic bins and a logarithmic axis solve different parts of the same plotting problem when data span several orders of magnitude.
6. A visible pattern is not automatically a physical law. Selection effects and measurement provenance are part of the result, not footnotes to ignore.

What this analysis can and cannot say

Supported by this analysis	Not established by this analysis
The archived Transit and RV samples have very different observed period distributions.	Transit and RV discoveries come from intrinsically separate natural planet populations.
The measured RV sample has a higher median reported mass than the measured Transit sample.	RV planets are universally more massive than Transit planets.
The mass histogram contains an apparent valley between lower-mass and giant-planet concentrations.	The valley is definitely caused by one particular planet-formation mechanism.
Selection effects and missing-measurement patterns materially affect the comparison.	Percentage normalisation removes all survey bias.

My strongest conclusion is therefore:

The planets represented in the NASA Exoplanet Archive look substantially different depending on whether they were discovered by Transit or Radial Velocity. Transit discoveries are concentrated at shorter periods, while the measured RV sample is shifted toward longer periods and larger reported masses. These are differences in the *observed archive samples*. They are shaped by detection methods, follow-up choices, missing measurements and the way mass is reported.

AI use declaration

I used ChatGPT as a tutor, brainstorming partner and code reviewer during this project.

It helped me generate possible analysis directions, understand choices such as duplicate filtering, percentage-normalised histograms and logarithmic bins, and check individual code snippets or error messages when I got stuck. I also used it to help organise and edit this write-up.

I downloaded the dataset, wrote and ran the Python analysis, inspected the outputs, chose which variables and plots to keep, and made the final decisions about interpretation.

I did not accept numerical results without reproducing them using my own code. The final responsibility for the analysis and its limitations is mine.

Data, code and sources

Tools used:

Python 3.11, pandas, NumPy and Matplotlib.

Data source:

NASA Exoplanet Archive, Planetary Systems table.

Archive DOI: [10.26133/NEA1](https://doi.org/10.26133/NEA1)

Useful documentation:

- [NASA Exoplanet Archive: Planetary Systems table and column definitions](#)
- [NASA Exoplanet Archive FAQ: default parameter sets](#)
- [NASA Science: how exoplanets are found and characterised](#)
- [NASA Science: what a transit is](#)

This is a learning project, not a peer-reviewed scientific study. The analysis reflects the downloaded archive snapshot used in July 2026 and the specific cleaning and plotting choices documented above.